

Categorical Data & Discrete Distributions

Categorical data

Contingency tables

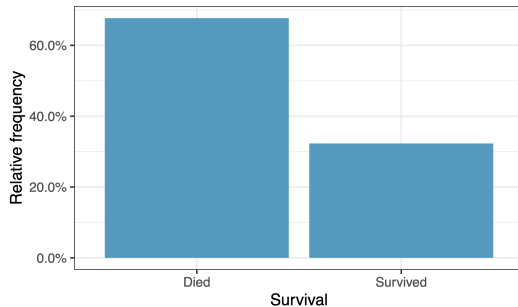
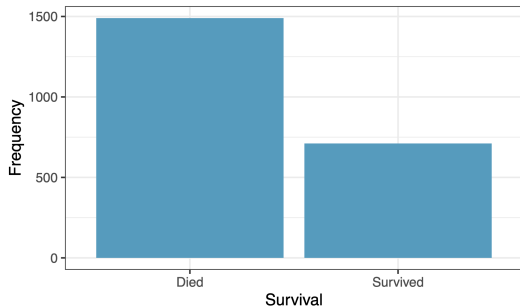
A table that summarizes data for two categorical variables is called a **contingency table**.

The contingency table below shows the distribution of survival and age of passengers on the Titanic.

		Died	Survived	Total
Age	Adult	1438	654	2092
	Child	52	57	109
	Total	1490	711	2201

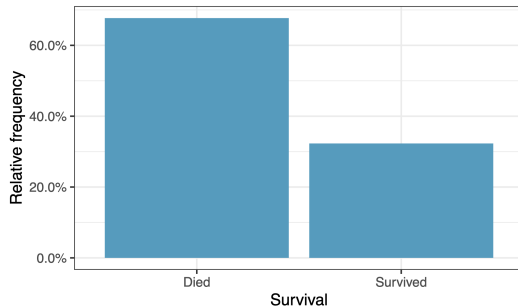
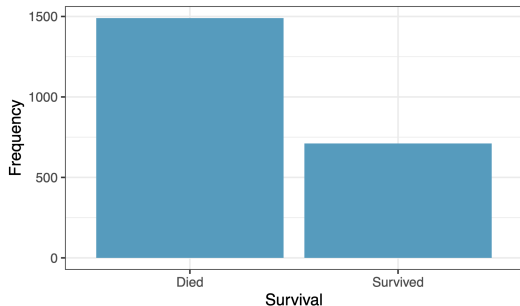
Bar plots

A **bar plot** is a common way to display a single categorical variable. A bar plot showing proportions instead of frequencies is called a **relative frequency bar plot**.



Bar plots

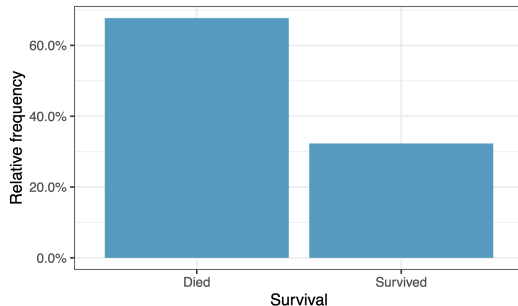
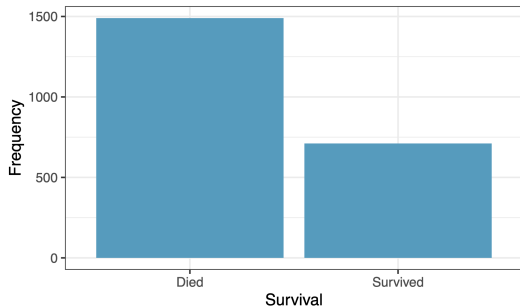
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How are bar plots different from histograms?

Bar plots display categorical variables; histograms display numerical variables. A histogram's x-axis is a number line, so bar order is fixed — a bar plot's categories can be reordered.

Choosing the appropriate proportion

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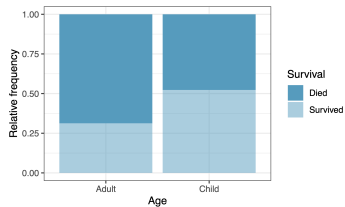
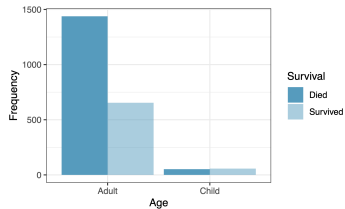
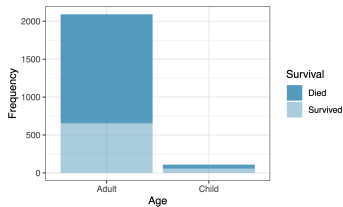
To answer this, we look at **row proportions**:

- ▶ % Adults who survived: $654/2092 \approx 0.31$
- ▶ % Children who survived: $57/109 \approx 0.52$

Bar plots with two variables

- ▶ **Stacked bar plot:** graphical display of contingency table counts, stacked on top of each other.
- ▶ **Side-by-side bar plot:** the same information, with bars placed next to each other instead of stacked.
- ▶ **Standardized (filled) bar plot:** graphical display of contingency table *proportions* rather than counts.

Comparing the three

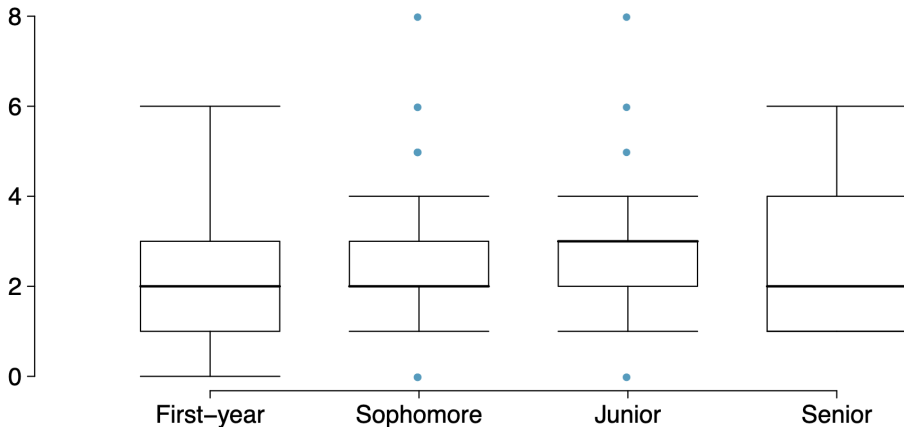


What are the differences between the three visualizations above? When would you reach for the standardized version instead of the stacked one?

Comparing a numerical variable across groups

Side-by-side boxplots let us compare a **numerical** variable across levels of a **categorical** variable.

Does there appear to be a relationship between class year and number of clubs students are in?



Discrete distributions

Random processes

- ▶ A **random process** is a situation where we know what outcomes could happen, but not which particular outcome will happen.
- ▶ Examples: coin tosses, die rolls, whether tomorrow's high temperature is above 70°F.
- ▶ It can be useful to *model* a process as random even if it isn't truly random (e.g., shuffle algorithms).

Probability basics

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Frequentist interpretation: the probability of an outcome is the proportion of times it would occur if we repeated the random process infinitely many times.

Disjoint vs. non-disjoint outcomes

Disjoint (mutually exclusive): cannot happen at the same time.

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- ▶ A single card drawn from a deck cannot be both an ace and a queen.

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Non-disjoint: can happen at the same time.

- ▶ A student can get an A in Stats and an A in Econ in the same semester.

Probability distributions

A **probability distribution** lists all possible events and the probability with which each occurs.

The distribution for the gender of one kid:

Event	Male	Female
Probability	0.5	0.5

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The distribution for the genders of two kids:

Event	MM	FF	MF	FM
Probability	0.25	0.25	0.25	0.25

Practice

Which of these variables would you expect to be uniformly distributed?

- a. weights of adult females
- b. salaries of a random sample of people from North Carolina
- c. house prices
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Answer: (d) — day-of-month birthdays have no reason to cluster around any particular value, unlike the other three, which are all right-skewed.

Random variables

- ▶ A **random variable** is a numeric quantity whose value depends on the outcome of a random event.
 - ▶ We use a capital letter, like X , to denote a random variable, and lowercase x for its possible values — e.g. $P(X = x)$.
- ▶ **Discrete random variables** often take only integer values.
 - ▶ Example: number of credit hours this term.
- ▶ **Continuous random variables** take real (decimal) values.
 - ▶ Example: cost of textbooks this term. (*We'll come back to continuous variables on Wednesday.*)

Expected value

We're often interested in the *average* outcome of a random variable — this is the **expected value** (mean), a weighted average of the possible outcomes:

$$\mu = E(X) = \sum_{i=1}^k x_i P(X = x_i)$$

Expected value: a card game

In a game of cards you win \$1 if you draw a heart, \$5 if you draw an ace (including the ace of hearts), \$10 if you draw the king of spades, and nothing for any other card. Write the probability model for your winnings, and calculate your expected winnings.

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Event	X	$P(X)$	$X \cdot P(X)$
Heart (not ace)	1	12/52	12/52
Ace	5	4/52	20/52
King of spades	10	1/52	10/52
All else	0	35/52	0
Total			$E(X) = 42/52 \approx 0.81$

So on average, you'd expect to win about 81 cents per game.